RFID Reader Development Guide

(C# Version)

V1.0

目录

1. Introduction 6
2. System Requirements 6
3. Installation & Uninstallation 6
4. Quick Start 6
4.1. SDK Development Workflow 6
4.2. Simple Usage Example 6
5. DeviceInstance Class 8
5.1. Event Description 8
5.2. Property Description 9
5.3. Method Description 9
5.3.1. Custructor 9
5.3.2. Open Device 10
5.3.3. Close Device 10
5.3.4. Check if Device Connection is Open 11
5.3.5. Check if Device Connection is Valid 11
5.3.6. Restart Reader 11
5.3.7. Reset Device 12
5.3.8. Query Device Parameter 12
5.3.9. Query RFID Parameter 12
5.3.10. Query UHF Parameter 13
5.3.11. Set Device Parameter 13
5.3.12. Set RFID Parameter 14
5.3.13. Set UHF Parameter 14
5.3.14. Query GPIO Mode 14
5.3.15. Set GPIO Mode 15
5.3.16. Set Single GPIO Mode 15
5.3.17. Query GPIO Level Status 15
5.3.18. Set GPIO Level Status 16
5.3.19. Set Single GPIO Level Status 16
5.3.20. Buzzer beeps 17
5.3.21. Start Asynchronous RFID Inventory 17
5.3.22. Synchronize RFID inventory 17
5.3.23. Stop RFID 18
5.3.24. Standing Wave Test 18
5.3.25. Read RFID tag data 18

5.3.26. Write RFID tag data 19
5.3.27. Lock RFID Tag 19
5.3.28. Kill RFID Tag 20
5.3.29. Firmware Upgrade 20
5.3.30. Get Serial Port Reader List 21
5.3.31. Get IP List 21
5.3.32. Start TCP Listen (Start the TCP server mode on the computer side) 21
5.3.33. Stop TCP Listen (Stop the TCP server mode on the computer side) 22
5.3.34. Create Listener Device Object 22
5.3.35. Scan WiFi hotspots 22
5.4. Enums 23
5.5. Device Class Enums 23
5.5.1. EDeviceGPIOIndex 23
5.5.2. EDeviceGPIOMode 23
5.5.3. EDeviceParam 23
5.5.4. EDeviceStatus 23
5.5.5. ENetworkMode 25
5.5.6. ESerialParity 25
5.5.7. EWiFiSecurityType 25
5.6. RFID Class Enums 26
5.6.1. ERFIDAntennaIndex 26
5.6.2. ERFIDAntennaMask 26
5.6.3. ERFIDParam 27
5.6.4. ERFIDStopReason 27
5.6.5. ERFIDTagType 27
5.7. UHF Class Enums 27
5.7.1. EUHFBank 27
5.7.2. EUHFParam 28
5.7.3. EUHFRegion 28
5.7.4. EUHFLockArea 28
5.7.5. EUHFLockType 28
6. Event Parameter 29
6.1. Device Event Parameters 29
6.1.1. DeviceConnectedEventArgs 29

6.1.2. DeviceDisconnectEventArgs 29
6.1.3. DeviceGPIOEventArgs 29
6.1.4. DeviceLogEventArgs 29
6.1.5. OTAEventArgs 30
6.2. RFID Class Event Parameter 30
6.2.1. RFIDStopEventArgs 30
6.2.2. RFIDTagArrivedEventArgs 30
7. Model 31
7.1. Device Class Model 31
7.1.1. DeviceGPIOModel 31
7.1.2. WiFiScanResultModel 31
7.2. RFID Class Model 31
7.2.1. RFIDInventoryModel 32
7.2.2. RFIDKillTagModel 33
7.2.3. RFIDLockTagModel 33
7.2.4. RFIDReadTagModel 34
7.2.5. RFIDTagModel 35
7.2.6. RFIDWriteTagModel 35
7.3. RFID Class Model 36
7.3.1. UHFSWRTestModel 36
8. Parameter 36
8.1. Parameter Classification 36
8.1.1. Device Class Parameter 36
8.1.2. RFID Class Parameter 38
8.1.3. UHF Class Parameter 40
8.2. Parameter Definition 40
8.2.1. ArrayModel 40
8.2.2. BitmapModel 41
8.2.3. BoolModel 41
8.2.4. ByteModel 41
8.2.5. BytesModel 41
8.2.6. DevicePwronModel 42
8.2.7. FlashModel 42
8.2.8. GPIOEventModel 42
8.2.9. IntModel 42
8.2.10. MaskModel 43

8.2.11. NetworkModel 45
8.2.12. PwrOnEventModel 45
8.2.13. RFIDAntennaModel 45
8.2.14. RFIDEASEventModel 46
8.2.15. RFIDIdleModel 46
8.2.16. RFIDKeyboardOutputModel 46
8.2.17. RFIDPowerModel 47
8.2.18. RFIDPropertyModel 47
8.2.19. RFIDProtocolModel 47
8.2.20. RFIDSpecialOutputModel 48
8.2.21. RFIDWiegandOuputModel 48
8.2.22. SerialModel 49
8.2.23. ShortModel 49
8.2.24. StringModel 49
8.2.25. TimestampModel 49
8.2.26. UHFEPCAlgoModel 50
8.2.27. UHFEPCBaseBandModel 50
8.2.28. UHFFrequencyModel 50
8.2.29. ParamModel 51
8.2.30. UHFSelectModel 51
8.2.31. VersionInfoModel 51
8.2.32. WiFiAPModel 52
8.2.33. WiFiConnectModel 52
Appendix 1 UHF Work Area Table 53

1. Introduction

The SDK is a specialized collection of APIs designed for our company's devices, aimed at assisting users in efficient and rapid application development for the equipment.

2. System Requirements

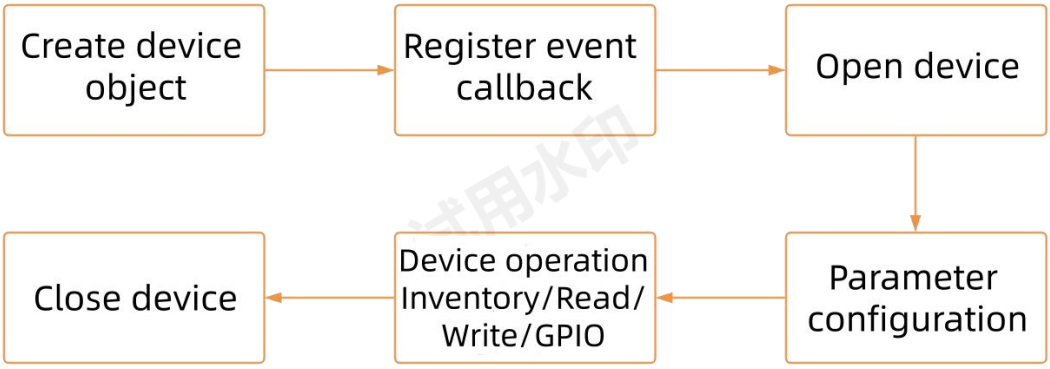
- **Operating System:** Windows 10/11
- **Processor:** Intel Core i3 or equivalent performance
- **Memory:** 8GB RAM (16GB recommended)
- **Storage Space:** At least 2GB free space
- **Other Dependencies:** .NET Framework 4.0

3. Installation & Uninstallation

The SDK is provided as a DLL and does not require installation.

4. Quick Start

4.1. SDK Development Workflow



4.2. Simple Usage Example

```
using DeviceAPI.Emums.RFID;
using DeviceAPI.Enums.Device;
using DeviceAPI.Instance;
using DeviceAPI.Models.Params;
using DeviceAPI.Utils;
using System;

namespace ConsoleDemo
{
 class BaseInventoryExample
 {
        //static String link = "COM2:115200";
        static String link = "192.168.1.201:8090";

        static void Main(string[] args)
        {
                DeviceInstance inst = new DeviceInstance();
                inst.RFIDTagArrivedEvent += Inst_RFIDTagArrivedEvent;
                inst.RFIDStopEvent += Inst_RFIDStopEvent;

                if (!inst.Open(link))
                {
                        Console.WriteLine($"Open link {link} failed!");
                        return;
                }
                Console.WriteLine($"Open link {link} success!");

                // enable antenna 1
                RFIDAntennaModel antennaModel = new RFIDAntennaModel();
                antennaModel.AntennaMask = 0x01;

                var rt = inst.SetRFIDParam(ERFIDParam.RFID_ANTENNA, antennaModel);
                if (rt != EDeviceStatus.SUCCESS)
                {
                        Console.WriteLine($"Set antenna enable failed: {rt}!");
                        return;
                }

                rt = inst.Inventory();
                if (rt != EDeviceStatus.SUCCESS)
                {
                        Console.WriteLine($"Start inventory failed: {rt}!");
                        return;
                }
                Console.WriteLine("Start inventory success");
```

```
            Console.WriteLine("Press any key to stop inventory");
            Console.ReadKey();

            inst.RFIDStop();
            Console.WriteLine("Stop inventory");
            inst.Close();
        }

        private static void Inst_RFIDStopEvent(object sender,
DeviceAPI.Events.RFID.RFIDStopEventArgs e)
        {
            Console.WriteLine($"RFIDStopEvent: Reason={e.Reason}");
        }

        private static void Inst_RFIDTagArrivedEvent(object sender,
DeviceAPI.Events.RFID.RFIDTagArrivedEventArgs e)
        {
            Console.WriteLine($"EPCTagArrivedEvent:
EPC={BytesConverter.ToHexString(e.Tag.EPC)}, Antenna={e.Tag.Antenna}");
        }

    }
  }
```

5. DeviceInstance Class

The DeviceInstance class encapsulates operations for the device. It covers opening/closing the device, querying/configuring parameters, GPIO control, device firmware upgrade, WiFi scanning, RFID inventory, RFID read/write tags, and other operations.

The DeviceInstance class is located in DeviceAPI.Instance.

5.1. Event Description

Event	Parameter Type	Description
ListenDeviceConnectedEvent	DeviceConnectedEventArgs	Event triggered when a

Event	Parameter Type	Description
		device connects to the SDK in server mode; Static property.
ListenDeviceLogEvent	DeviceLogEventArgs	SDK log event in server mode; Static property.
DeviceLogEvent	DeviceLogEventArgs	SDK Log Event
DeviceReconnectedEvent	DeviceConnectedEventArgs	Device reconnection event (TCP).
DeviceDisconnectEvent	DeviceDisconnectEventArgs	Device disconnection event.
RFIDTagArrivedEvent	RFIDTagArrivedEventArgs	Device disconnection event.
RFIDStopEvent	RFIDStopEventArgs	RFID stop event.
DeviceGPIOEvent	DeviceGPIOEventArgs	Device GPIO change event.
OTAEvent	OTAEventArgs	Firmware upgrade progress notification event.

Note: To improve SDK processing efficiency, avoid performing long business operations within the event handler function. Users should only receive data and synchronize it to other threads for processing.

5.2. Property Description

Proper	Type	Description
Addr	byte	Communication address, range 1~250; 255 is the broadcast address.
Link	string	Communication link name.
InListen	bool	Indicates whether currently in TCP server mode (listening mode); Static property.

5.3. Method Description

5.3.1. Custructor

	Tpye	Description
public DeviceInstance()		
Parameter		
None		

Return Value		
None		
Description		
Creates a DeviceInstance instance.		

5.3.2. Open Device

	Type	Description
public bool Open(string param, byte addr = BaseFrame.BROADCAST)		
Parameter		
param	string	Connection parameters: Serial connection: [COM Port:Baud Rate] TCP connection: [IP:Port] e.g., Serial: COM1:115200 TCP: 192.168.1.201:8090
addr	byte	Communication address, valid range 1~250, default is broadcast address.
Return Value		
	bool	Initialization success or failure.
Description		
Initializes the device, i.e., opens the device connection and performs necessary initialization operations.		

5.3.3. Close Device

	Type	Description
public void Close()		
Parameter		
None		
Return Value		
None		
Description		
Releases device resources and disconnects the device.		

5.3.4. Check if Device Connection is Open

	Type	Description
public bool IsConnected()		
Parameter		
None		
Return Value		
	bool	Return whether the device connection is open.
Description		
Only checks if the current connection is open, does not perform communication testing.		

5.3.5. Check if Device Connection is Valid

	Type	Description
public EDeviceStatus CheckConnect()		
Parameter		
None		
Return Value		
	EDeviceStatus	Return the device connection check result.
Description		
Checks if the current connection is valid by performing a communication test.		

5.3.6. Restart Reader

	Type	Description
public EDeviceStatus Restart()		
Parameter		
None		
Return Value		
	EDeviceStatus	Return the restart result.
Description		
Restarts the reader.		

5.3.7. Reset Device

	Type	Description
public EDeviceStatus Restore()		
Parameter		
None		
Return Value		
	EDeviceStatus	Return the device reset result.
Description		
Resets the device (restores factory settings).		

5.3.8. Query Device Parameter

	Type	Description
public EDeviceStatus GetDeviceParam<T>(EDeviceParam id, T val)		
Parameter		
id	EDeviceParam	Device ParameterType
val	ParamModel	Device parameter object used to store the queried parameter value. Data type must match the parameter type.
Return Value		
	EDeviceStatus	Return the query result.
Description		
Queries device parameters. Device parameter definitions and value types are detailed in the Device Parameters chapter.		

5.3.9. Query RFID Parameter

	Type	Description
public EDeviceStatus GetRFIDParam<T>(ERFIDParam id, T val)		
Parameter		
id	ERFIDParam	RFIDParameterType
val	ParamModel	RFID parameter object used to store the queried parameter value. Data type must match the parameter type.
Return Value		
	EDeviceStatus	Return the query result.

Description
Queries RFID parameters. RFID parameter definitions and value types are detailed in the RFID Parameters chapter.

5.3.10. Query UHF Parameter

	Type	Description
public EDeviceStatus GetUHFParam<T>(EUHFParam id, T val)		
Parameter		
id	EUHFParam	UHFParameterType
val	ParamModel	UHF parameter object used to store the queried parameter value. Data type must match the parameter type.
Return Value		
	EDeviceStatus	Return the query result.
Description		
Queries UHF parameters. UHF parameter definitions and value types are detailed in the UHF Parameters chapter.		

5.3.11. Set Device Parameter

	Type	Description
public EDeviceStatus SetDeviceParam<T>(EDeviceParam id, T val, bool saved = true)		
Parameter		
key	EDeviceParam	Device ParameterType
val	ParamModel	Device parameter object storing the parameter value to be set. Data type must match the parameter type.
saved	bool	Is it permanently stored (stored in device Flash)
Return Value		
	EDeviceStatus	Return the setting result.
Description		
Sets device parameters. Device parameter definitions and value types are detailed in the Device Parameters chapter.		

5.3.12. Set RFID Parameter

	Type	Description
public EDeviceStatus SetRFIDParam<T>(ERFIDParam id, T val, bool saved = true)		
Parameter		
key	ERFIDParam	RFIDParameterType
val	ParamModel	RFID parameter object storing the parameter value to be set. Data type must match the parameter type.
saved	bool	Is it permanently stored (stored in device Flash)
Return Value		
	EDeviceStatus	Return the setting result.
Description		
Sets RFID parameters. RFID parameter definitions and value types are detailed in the RFID Parameters chapter.		

5.3.13. Set UHF Parameter

	Type	Description
public EDeviceStatus SetUHFParam<T>(EUHFParam id, T val, bool saved = true)		
Parameter		
key	EUHFParam	UHFParameterType
val	ParamModel	UHF parameter object storing the parameter value to be set. Data type must match the parameter type.
saved	bool	Is it permanently stored (stored in device Flash)
Return Value		
	EDeviceStatus	Return the setting result.
Description		
Sets UHF parameters. UHF parameter definitions and value types are detailed in the UHF Parameters chapter.		

5.3.14. Query GPIO Mode

	Type	Description
public EDeviceStatus GetGPIOMode(Dictionary<EDeviceGPIOIndex, EDeviceGPIOMode> mode)		
Parameter		
mode	Dictionary<EDeviceGPIOIndex,	Used to store the queried GPIO

	EDeviceGPIOMode>	mode results.
Return Value		
	EDeviceStatus	Return the query result.
Description		
Queries the reader GPIO mode, returned as a dictionary.		

5.3.15. Set GPIO Mode

	Type	Description
public EDeviceStatus SetGPIOMode(Dictionary<EDeviceGPIOIndex, EDeviceGPIOMode> mode)		
Parameter		
mode	Dictionary<EDeviceGPIOIndex, EDeviceGPIOMode>	Dictionary containing the GPIO modes to be set.
Return Value		
	EDeviceStatus	Return the setting result.
Description		
Sets the reader GPO output, parameters are passed as a dictionary		

5.3.16. Set Single GPIO Mode

	Type	Description
public EDeviceStatus SetGPIOMode(EDeviceGPIOIndex gpio, EDeviceGPIOMode mode)		
Parameter		
gpio	EDeviceGPIOIndex	GPIO index that needs to be set
mode	EDeviceGPIOMode	GPIO Mode
Return Value		
	EDeviceStatus	Return the setting result.
Description		
Set device GPIO mode		

5.3.17. Query GPIO Level Status

	Type	Description
public EDeviceStatus GetGPIOState(Dictionary<EDeviceGPIOIndex, bool> state)		

Parameter		
state	Dictionary<EDeviceGPIOIndex, bool>	Used to store GPIO level status results for queries
Return Value		
	EDeviceStatus	Return the query result.
Description		
Query the GPIO status of the device and return it in the form of a dictionary. Return Valuetrue is high and false is low		

5.3.18. Set GPIO Level Status

	Type	Description
public EDeviceStatus SetGPIOState(Dictionary<EDeviceGPIOIndex, bool> state)		
Parameter		
state	Dictionary<EDeviceGPIOIndex, bool>	Used to store GPIO level state dictionary that needs to be set
Return Value		
	EDeviceStatus	Return the setting result
Description		
Set the GPIO level status of the device and pass the Parameter in dictionary form, with a value of true indicating high level and false indicating low level		

5.3.19. Set Single GPIO Level Status

	Type	Description
public EDeviceStatus SetGPIOState(EDeviceGPIOIndex gpio, bool state)		
Parameter		
gpio	EDeviceGPIOIndex	GPIO index that needs to be set
state	bool	GPIO Level
Return Value		
	EDeviceStatus	Return the setting result
Description		
Set the GPIO level status of the device		

5.3.20. Buzzer beeps

	Type	Description
public EDeviceStatus Beep(int time = 50)		
Parameter		
time	int	Buzzer duration, unit: ms
Return Value		
	EDeviceStatus	Return the setting result
Description		
The buzzer rings once		

5.3.21. Start Asynchronous RFID Inventory

	Type	Description
public EDeviceStatus Inventory(RFIDInventoryModel model = null)		
Parameter		
model	RFIDInventoryModel	Iventory Parameter
Return Value		
	EDeviceStatus	Return the result of opening asynchronous inventory
Description		
The device performs asynchronous RFID inventory. After successful activation, the tags counted by the device are transmitted to the application through RFID tag data arrival events		

5.3.22. Synchronize RFID inventory

	Type	Description
public EDeviceStatus RFIDInventory(int milliseconds, List<RFIDTagModel> list, RFIDInventoryModel model = null)		
Parameter		
milliseconds	int	Inventory duration, in milliseconds (ms)
list	List<RFIDTagModel>	List for storing label data
model	RFIDInventoryModel	Iventory Parameter
Return Value		
	EDeviceStatus	Return synchronized

		inventory results
Description		
Perform RFID synchronous inventory of equipment. After the inventory is completed, the data collected will be merged and returned through a list. Receiving tag data during the inventory process will not trigger an RFID tag data arrival event.		

5.3.23. Stop RFID

	Type	Description
public EDeviceStatus RFIDStop()		
Parameter		
None		
Return Value		
	EDeviceStatus	Return the stop result
Description		
Used to stop RFID operations. Manual stop commonly used for asynchronous inventory		

5.3.24. Standing Wave Test

	Type	Description
public EDeviceStatus UHFSWRTest(UHFSWRTestModel model)		
Parameter		
model	UHFSWRTestModel	Parameters for standing wave testing, transmitting the parameters and results of standing wave testing
Return Value		
	EDeviceStatus	Return test results
Description		
Perform standing wave measurements on designated antennas, areas, and channels		

5.3.25. Read RFID tag data

	Type	Description
public EDeviceStatus RFIDReadTag(RFIDReadTagModel model)		

Parameter		
model	RFIDReadTagModel	Read RFID tag parameters, transmit tag read parameters and read back data
Return Value		
	EDeviceStatus	Return the result of RFID tag reading operation
Description		
Read RFID tags according to the specified parameters, and after reading, the read data will be returned through the model. When using the tag operation interface to operate a specified tag, first call the UHFParameter interface to set TAG-SELECTn to match the specified tag, and then call the tag operation interface to perform the operation.		

5.3.26. Write RFID tag data

	Type	Description
public EDeviceStatus RFIDWriteTag(RFIDWriteTagModel model)		
Parameter		
model	RFIDWriteTagModel	Write RFID tag parameters, transmit the parameters and data written by the tag
Return Value		
	EDeviceStatus	Return the write tag operation result.
Description		
Write RFID tags according to the specified parameters. When using the tag operation interface to operate a specified tag, first call the UHFParameter interface to set TAG-SELECTn to match the specified tag, and then call the tag operation interface to perform the operation.		

5.3.27. Lock RFID Tag

	Type	Description
public EDeviceStatus RFIDLockTag(RFIDLockTagModel model)		
Parameter		

model	RFIDLockTagModel	Lock RFID tag parameters
Return Value		
	EDeviceStatus	Return the result of the lock label operation
Description		
Lock RFID tags according to specified parameters. When using the tag operation interface to operate a specified tag, first call the UHFParameter interface to set TAG-SELECTn to match the specified tag, and then call the tag operation interface to perform the operation.		

5.3.28. Kill RFID Tag

	Type	Description
public EDeviceStatus RFIDKillTag(RFIDKillTagModel model)		
Parameter		
model	RFIDKillTagModel	Kill tag parameters.
Return Value		
	EDeviceStatus	Return the kill tag operation result.
Description		
Perform tag killing operation according to the specified parameter. When using the tag operation interface to operate a specified tag, first call the UHFParameter interface to set TAG-SELECTn to match the specified tag, and then call the tag operation interface to perform the operation.		

5.3.29. Firmware Upgrade

	Type	Description
public EDeviceStatus OTA(byte[] firmware)		
Parameter		
firmware	byte[]	Firmware data.
Return Value		
	EDeviceStatus	Return the firmware upgrade result.
Description		
Firmware upgrade must use the firmware version corresponding to the reader hardware. Incorrect firmware may cause the reader to fail to start.		

5.3.30. Get Serial Port Reader List

	Type	Description
public static List<string> GetSerialPortList()		
Parameter		
None		
Return Value		
	List<string>	Return the list of serial port readers.
Description		
Used to query the list of serial ports on the computer.		

5.3.31. Get IP List

	Type	Description
public static List<string> GetTCPIpList()		
Parameter		
None		
Return Value		
	List<string>	Return the list of scanned reader IPs.
Description		
Used to query the list of scanned reader IPs on the computer.		

5.3.32. Start TCP Listen (Start the TCP server mode on the computer side)

	Type	Description
public static bool StartListen(string param)		
Parameter		
param	string	Binding Parameter, e.g., 192.168.1.99:8091
Return Value		
	bool	Return the TCP listen start result.
Description		
Used to start a TCP listening service, allowing readers configured in active connection server mode to actively connect to the software. After a reader connects, the ListenReaderConnectingEvent is triggered. The user needs to call the Create Listener Reader Object (BuildListenReader) interface to		

generate a new RFIDReader object.

5.3.33. Stop TCP Listen (Stop the TCP server mode on the computer side)

	Type	Description
public static void StopListen()		
Parameter		
None		
Return Value		
None		
Description		
Used to stop the TCP listening service.		

5.3.34. Create Listener Device Object

	Type	Description
public static DeviceInstance BuildListenInstance(String link)		
Parameter		
link	String	Communication link name.
Return Value		
	DeviceInstance	Device object.
Description		
Create a DeviceInstance object for the monitored device		

5.3.35. Scan WiFi hotspots

	Type	Description
public List<WiFiScanResultModel> ScanWiFi(uint timout = 20000)		
Parameter		
timout	uint	Scan timeout, unit : ms
Return Value		
	List<WiFiScanResultModel>	WiFi scan result list
Description		
Search for WiFi hotspots		

5.4. Enums

5.5. Device Class Enums

The device class enums is located at DeviceAPI.Enum.Device.

5.5.1. EDeviceGPIOIndex

GPIO number, used for GPIO operations, GPIO triggering, etc

Definition	Description
GPIO1	GPIO1 number
GPIO2	GPIO2 number
GPIO3	GPIO3 number
GPIO3	GPIO3 number
GPIO4	GPIO4 number
GPIO5	GPIO5 number
GPIO6	GPIO6 number
GPIO7	GPIO7 number

5.5.2. EDeviceGPIOMode

GPIO Mode

Definition	Description
OUTPUT	GPIO output mode
INPUT	GPIO input mode

5.5.3. EDeviceParam

Device Parameter enums, see Device Parameter section.

5.5.4. EDeviceStatus

Device status code

Definition	Description
SUCCESS	Success
COMMAND_INVALID_TYPE	Invalid command Type
COMMAND_INVALID_LENGTH	Invalid command Length
COMMAND_INVALID_PARAM	Invalid command Parameter
COMMAND_EXEC_ERROR	Command execution error
COMMAND_NO_RESPONSE	Command no response
GPIO_OVER_RANGE	GPIO Index Over range
GPIO_INVALID_MODE	Invalid GPIO Mode
GPIO_OTHER_ERROR	Other GPIO errors
PARAM_INVALID_TYPE	Invalid Parameter Type
PARAM_INVALID_VALUE	Invalid Parameter Value
PARAM_INVALID_LENGTH	Invalid Parameter Length
PARAM_GET_FAILED	Failed to get Parameter
PARAM_SET_FAILED	Failed to set Parameter
PARAM_SAVE_FAILED	Failded to save Parameter
PARAM_OTHER_ERROR	Other Parameter errors
TAG_OTHER_ERROR	Tag returns other errors
TAG_NOT_SUPPORT	Tag return is not supported
TAG_INSUFFICIENT_PRIVILEGES	Insufficient tag return permission
TAG_MEMORY_OVERRUN	Tag returns memory overflow
TAG_MEMORY_LOCKED	Tag returns memory locked
TAG_CRYPTOGRAPHIC_SUITE_ERROR	Tag return encryption suite error
TAG_COMMAND_NOT_ENCAPSULATED	Tag return command is not encapsulated
TAG_RESPONSEBUFFER_OVERFLOW	Tag return response buffer overflow
TAG_SECURITY_TIMEOUT	Tag return security timeout
TAG_INSUFFICIENT_POWER	Insufficient tag return power
TAG_LOST	Tag lost
TAG_NON_SPECIFIC_ERROR	Non-specific error
RFID_INVALID_REGION	Invalid RFID Region
RFID_INVALID_VALUE	Invalid RFID Value
RFID_EXEC_ERROR	RFID execution error
RFID_NO_ACK	RFID no response

Definition	Description
RFID_INTERNEL_ERROR	RFID internal error
RFID_OTHER_ERROR	Other RFIDParameter
DEVICE_BUSY	Device busy
INTERNAL_ERROR	Internal error
INVALID_LINK	Invalid link

5.5.5. ENetworkMode

Network mode

Definition	Description
TCP_SERVER	TCP Server Mode
TCP_CLIENT	TCP Client Mode
UDP_SERVER	UDP Server Mode
UDP_CLIENT	UDP Client Mode

5.5.6. ESerialParity

Serial port parity bit

Definition	Description
NONE	No parity
ODD	Odd parity
EVEN	Even parity

5.5.7. EWiFiSecurityType

WiFi hotspot security type

Definition	Description
Open	Open
WEP	WEP
WPA_TKIP	WPA TKIP
WPA_AES	WPA AES
WPA_Mixed	WPA Mixed

WPA2_AES	WPA2 AES
WPA2_TKIP	WPA2 TKIP
WPA2_Mixed	WPA2 Mixed
WPA_WPA2_TKIP	WPA/WPA2 TKIP
WPA_WPA2_AES	WPA/WPA2 AES
WPA_WPA2_Mixed	WPA/WPA2 Mixed
WPA2_Enterprise	WPA2 Enterprise
WPA_WPA2_Enterprise	WPA/WPA2 Enterprise
WPA3_ASE_AES	WPA3-ASE AES
UnknownType	Unknown Type

5.6. RFID Class Enums

The device class enums is located in DeviceAPI.Enum.RFID.

5.6.1. ERFIDAntennaIndex

Antenna number, used for carrier testing, tag data, etc

Definition	Description
ANT1	Antenna 1
ANT2	Antenna 2
...	...
ANT16	Antenna 16

5.6.2. ERFIDAntennaMask

Antenna mask, used for line configuration

Definition	Description
ANT1	Antenna 1
ANT2	Antenna 2
...	...
ANT16	Antenna 16

5.6.3. ERFIDParam

RFID Parameter enums, see RFIDParameter chapter.

5.6.4. ERFIDStopReason

RFID Stop Reason

Definition	Description
FINISHED	Proactively end
STOPED	Passive stop
ERROR	Abnormal stop

5.6.5. ERFIDTagType

RFID Stop Reason

Definition	Description
ISO18K6B	ISO18000-6B Protocal
ISO18K6C	ISO18000-6C Protocal
ISO18K6D	ISO18000-6D Protocal

5.7. UHF Class Enums

UHF Class Enums is located in DeviceAPI.Enum.UHF。

5.7.1. EUHFBank

EPC tag storage area, used for operations such as reading and writing tags

Definition	Description
ISO18K6C_RESERVED	6C Tag Reserve Area

ISO18K6C_EPC	6C Tag EPC area
ISO18K6C_TID	6C Tag TID area
ISO18K6C_USER	6C Tag User area

5.7.2. EUHFParam

UHFParameter Enums， see UHFParameter chapter .

5.7.3. EUHFRegion

UHF working area, refer to the UHF working area table chapter.

5.7.4. EUHFLockArea

UHF tag lock area, used for tag locking operations

定义	Description
ISO18K6C_User	6C Tag User area
ISO18K6C_TID	6C Tag TID area
ISO18K6C_EPC	6C Tag EPC area
ISO18K6C_AccessPasswd	6C Tag access password
ISO18K6C_KillPasswd	6C Tag kill password

5.7.5. EUHFLockType

UHF tag lock type, used for tag locking operations

Definition	Description
ReadWrite	Readable and writable
AlwasyReadWrite	Permanently writable
PasswdWrite	Can be password written
AlwaysReadOnly	Forever not writable

6. Event Parameter

6.1. Device Event Parameters

The device event parameter is located in DeviceAPI.Events. Device

6.1.1. DeviceConnectedEventArgs

Device connection success event

Attribute	Type	Description
Link	string	Communication link name

6.1.2. DeviceDisconnectEventArgs

Device disconnection event Parameter

Attribute	Type	Description
Link	string	Communication link name
WillRetry	bool	Will the SDK retry the connection in the background

6.1.3. DeviceGPIOEventArgs

GPIO level change event Parameter

Attribute	Type	Description
Link	string	Communication link name
GPI	DeviceGPIOModel	GPIO Level status data

6.1.4. DeviceLogEventArgs

Log events Parameter

Attribute	Type	Description

Link	string	Communication link name
Message	String	log message

6.1.5. OTAEventArgs

OTA Event Parameter

Attribute	Type	Description
Link	string	Communication link name
Percent	float	Upgrade progress percentage，0~100

6.2. RFID Class Event Parameter

The RFID class event parameter is located in DeviceAPI.Events.RFID.

6.2.1. RFIDStopEventArgs

RFID Stop Event Parameter

Attribute	Type	Description
Link	string	Communication link name
Reason	ERFIDStopReason	RFID Stop reason

6.2.2. RFIDTagArrivedEventArgs

Received tag data event Parameter

Attribute	Type	Description
Link	string	Communication link name
Tag	RFIDTagModel	RFID Tag data

7. Model

7.1. Device Class Model

Device Class Model is located in DeviceAPI.Models.Device。

7.1.1. DeviceGPIOModel

GPIO level state change data

Attribute	Type	Description
Index	EDeviceGPIOIndex	GPIO Index
Level	bool	GPIO Level status
Timestamp	DateTime	Time stamp

7.1.2. WiFiScanResultModel

WiFi hotspot scanning results

Attribute	Type	Description
Id	short	WiFI 热点索引
SSID	String	WiFI 热点 SSID
BSSID	byte[]	WiFi 热点的 BSSID
RSSI	short	WiFi 热点的 RSSI
Chnnael	short	WiFi 热点使用的信道
SecurityType	EWiFiSecurityType	WiFi 热点使用的安全 Type

7.2. RFID Class Model

RFID Class Model is located DeviceAPI.Models.RFID。

7.2.1. RFIDInventoryModel

Inventory RFID tag parameters

Attribute	Type	Description
RequestTID	bool	Request to read TID, default value false
RequestFrequency	bool	Request transmission frequency, default false
RequestPhase	bool	Request transmission phase, default value false
RequestCount	bool	Request transmission read times, default false
RequestTimestamp	bool	Request transmission timestamp, default value false
RequestFrequency	bool	Request transmission frequency, default false
RequestRSSI	bool	Request to transmit RSSI, default false
EnableInventoryTime	bool	Enable inventory duration option, default false
InventoryTime	uint	Inventory duration, in mS. 0 indicates continuous inventory until receiving the StopRFID instruction. The default value is 0.
EnableTagSelect	bool	Enable tag selection option, default false
SelectEnableMask	uint	Tag selection enable mask， bit0- select 1, bit1- select 2...
EnablePassword	bool	Enable password option, default to false
Password	byte[]	Password
EnablePresetPassword	bool	Enable preset password option, default false
PresetPasswordIndex	int	Preset password index
EnableTemperatureTag	bool	Enable temperature label option, default false
TemperatureTagType	uint	Temperature Tag Type
TemperatureTagArgs	uint	Temperature Tag Parameter
EnableHumilityTag	bool	Enable humidity label option, default false
HumilityTagArgs	uint	Humility Tag Type
TemperatureTagArgs	uint	Humility Tag Parameter
EnableLEDTag	bool	Enable LED label option, default false
LEDTagType	uint	Temperature LEDType
LEDTagArgs	uint	Temperature LEDParameter
EnableImpinjTag	bool	Enable ImpinJ tag option, default false
EnableTagFocus	bool	Enable TagFocus, default value false
EnableFastID	bool	Enable FastID, default value false

EnableGen2X	bool	Enable G2X, default value false
EnableTagRead	bool	Enable tag reading, default value false
ReadBank	short	Tag reading area
ReadWordAddress	short	Tag reading word address
ReadWordLength	short	Tag reading word length
EnableTagFitler	bool	Enable tag filtering (discard tags with RSSI less than the specified value), default to false
RSSIFilter	float	RSSI filtering value, unit dBm
EnableTagMerge	bool	Enable merge reporting, default false
TagMergeTime	uint	Tag merging time, in milliseconds
EnableTagCache	bool	Merge reporting, default false

7.2.2. RFIDKillTagModel

Kill RFID tag parameters

Attribute	Type	Description
EnableKillTimeout	bool	Enable duration option, default false
KillTimeout	uint	Duration, unit: ms. “0” indicates a timeout for device self-management. The default value is “0”.
EnableTagSelect	bool	Enable tag selection option, default false
SelectEnableMask	uint	Tag selection enable mask, bit0- select 1, bit1- select 2...
EnablePassword	bool	Enable password option, default to false
Password	byte[]	Password
EnablePresetPassword	bool	Enable preset password option, default false
PresetPasswordIndex	int	Preset password index

7.2.3. RFIDLockTagModel

Lock RFID Tag Parameter

Attribute	Type	Description
EnableLockTimeout	bool	Enable duration option, default false
LockTimeout	uint	Duration, unit: ms. “0” indicates a

		timeout for device self-management. The default value is 0.
EnableTagSelect	bool	Enable tag selection option, default false
SelectEnableMask	uint	Tag selection enable mask，bit0- select 1, bit1- select 2...
EnablePassword	bool	Enable password option, default to false
Password	byte[]	Password
EnablePresetPassword	bool	Enable preset password option, default false
PresetPasswordIndex	int	Preset password index
LockArea	short	Lock Area
LockType	short	Lock Type

7.2.4. RFIDReadTagModel

Read Tag Parameter

Attribute	Type	Description
EnableReadTimeout	bool	Enable duration option, default false
ReadTimeout	uint	Duration, unit: ms. “0” indicates a timeout for device self-management. The default value is “0”.
EnableTagSelect	bool	Enable tag selection option, default false
SelectEnableMask	uint	Tag selection enable mask，bit0- select 1, bit1- select 2...
EnablePassword	bool	Enable password option, default to false
Password	byte[]	Password
EnablePresetPassword	bool	Enable preset password option, default false
PresetPasswordIndex	int	Preset password index
Bank	short	Read area
WordStart	short	Read word address
WordLength	shot	Read word length
ReadBackData	byte[]	Read data

7.2.5. RFIDTagModel

RFID Tag Data

Attribute	Type	Description
TagType	ERFIDTagType	Tag Type
PC	ushort	PC Data
EPC	byte[]	EPC Data
TID	byte[]	TID Data
Data	byte[]	Data read during inventory
Antenna	ERFIDAntennaIndex	Read the antenna index of the tag
Count	uint	Read count
RSSI	float	Tag RSSI value, unit dBm
Frequency	uint	Tag reading frequency, Unit: kHz
Phase	short	Tag reading phase 0~360
Timestamp	DateTime	Time stamp
Temperature	float	Tag temperature value, unit ℃
Humidity	float	Tag humidity value, unit:%

7.2.6. RFIDWriteTagModel

Write RFID Tag Parameter

Attribute	Type	Description
EnableWriteTimeout	bool	Enable duration option, default false
WriteTimeout	uint	Duration, unit: ms. “0” indicates a timeout for device self-management. The default value is “0”.
EnableTagSelect	bool	Enable tag selection option, default false
SelectEnableMask	uint	Tag selection enable mask, bit0- select 1, bit1- select 2...
EnablePassword	bool	Enable password option, default to false
Password	byte[]	Password
EnablePresetPassword	bool	Enable preset password option, default false
PresetPasswordIndex	int	Preset password index
Bank	short	Write Area
BLockSize	short	Block Size

WordStart	short	Write word address
WordLength	shot	Write word length
Data	byte[]	Written data

7.3. RFID Class Model

RFID Class Model is located in DeviceAPI.Models.RFID。

7.3.1. UHFSWRTestModel

UHF Standing wave test Parameter

Attribute	Type	Description
Antenna	ERFIDAntennaIndex	Antenna number for carrier testing, default antenna 1
Region	EUHFRegion	Work area, see area list
FrequencyKHz	int	Test frequency, unit: kHz
Power	short	The power used for standing wave testing, unit : 0.01dBm
RL	short	Return the return loss obtained from the test, unit: 0.01dBm

8. Parameter

8.1. Parameter Classification

8.1.1. Device Class Parameter

Parameter Definition	Parameter Name	Read Write	ParameterType
DEVICE_VERSION	Motherboard version information	RO	VersionInfoModel
DEVICE_SN	Motherboard serial number	RO	StringModel

Parameter Definition	Parameter Name	Read Write	ParameterType
ETHERNET_MAC	Ethernet MAC	RO	BytesModel
WIFI_MAC	WIFI MAC	RO	BytesModel
BT_MAC	BT_MAC	RO	BytesModel
DEVICE_NAME	Device Name	RW	StringModel
DEVICE_ASSET_NO	Asset Number	RW	StringModel
DEVICE_DATETIME	Device Date	RW	TimestampModel
DEVICE_PWRON_INFO	Device Power-on Information	RO	DevicePwronModel
PWRON_EVENT	Power-on Event	RW	PwrOnEventModel
DEVICE_TEMPERATRUE	Device Temperature	RO	ShortModel
DEVICE_SOC	Device SOC	RO	ShortModel
DEVICE_ADDRESS	Device Address	RW	ByteModel
RS232_ATTR	RS232Attribute	RW	SerialModel
RS485_ATTR	RS485Attribute	RW	SerialModel
ETHERNET_NETWORK	Ethernet network Attribute	RW	NetworkModel
WIFI_NETWORK	WiFi network Attribute	RW	NetworkModel
WIFI_AP_ATTR	WiFi connection AP	RW	WiFiAPModel
WIFI_CONNECT_INFO	WiFi connection information	RO	WiFiConnectModel
DEFAULT_LINK	Default Connection	RW	ByteModel
BUZZER_ATTR	Buzzer Parameter	RW	FlashModel
LED_COMM_ATTR	Communication indicator light Parameter	RW	FlashModel
LED_TAG_ATTR	Tag indicator light Parameter	RW	FlashModel
GPIO1_EVENT	GPIO1 Interrupt triggering event	RW	GPIOEventModel
GPIO2_EVENT	GPIO2 Interrupt triggering event	RW	GPIOEventModel
GPIO3_EVENT	GPIO3 Interrupt triggering event	RW	GPIOEventModel
GPIO4_EVENT	GPIO4 Interrupt	RW	GPIOEventModel

Parameter Definition	Parameter Name	Read Write	ParameterType
	triggering event		
GPIO5_EVENT	GPIO5 Interrupt triggering event	RW	GPIOEventModel
GPIO6_EVENT	GPIO6 Interrupt triggering event	RW	GPIOEventModel
GPIO7_EVENT	GPIO7 Interrupt triggering event	RW	GPIOEventModel
GPIO8_EVENT	GPIO8 Interrupt triggering event	RW	GPIOEventModel

8.1.2. RFID Class Parameter

Parameter Definition	Parameter Name	Read Write	ParameterType
RFID_VERSION	RFID Version Information	RO	VersionInfoModel
RFID_SN	RFID SN	RO	StringModel
RFID_PROPERTY	RFIDAttribute	RO	RFIDPropertyModel
RFID_PROTOCOL	RFID Current Protocal	RW	IntModel
RFID_TEMPERATURE	RFID Temperature	RO	ShortModel
RFID_IDLE	RFID Idle Mode	RW	RFIDIdleModel
RFID_EAS	RFID EAS Event	RW	RFIDEASEventModel
WIEGAND_OUTPUT	Wiegand Output	RW	RFIDWiegandOuputModel
SPECIAL_OUTPUT	Special format output	RW	RFIDSpecialOutputModel
KEYBOARD_OUTPUT	Special format output	RW	RFIDKeyBoardOutputModel
RFID_ANTENNA_STATE	RFID Antenna Status	RO	MaskModel
RFID_ANTENNA	Antenna Configuration	RW	RFIDAntennaModel
RFID_ANTENNA_POWER	All antenna power configurations	WO	RFIDPowerModel

Parameter Definition	Parameter Name	Read Write	ParameterType
RFID_ANT1_POWER	Antenna 1 Power	RW	RFIDPowerModel
RFID_ANT2_POWER	Antenna 2 Power	RW	RFIDPowerModel
RFID_ANT3_POWER	Antenna 3 Power	RW	RFIDPowerModel
RFID_ANT4_POWER	Antenna 4 Power	RW	RFIDPowerModel
RFID_ANT5_POWER	Antenna 5 Power	RW	RFIDPowerModel
RFID_ANT6_POWER	Antenna 6 Power	RW	RFIDPowerModel
RFID_ANT7_POWER	Antenna 7 Power	RW	RFIDPowerModel
RFID_ANT8_POWER	Antenna 8 Power	RW	RFIDPowerModel
RFID_ANT9_POWER	Antenna 9 Power	RW	RFIDPowerModel
RFID_ANT10_POWER	Antenna 10 Power	RW	RFIDPowerModel
RFID_ANT11_POWER	Antenna 11 Power	RW	RFIDPowerModel
RFID_ANT12_POWER	Antenna 12 Power	RW	RFIDPowerModel
RFID_ANT13_POWER	Antenna 13 Power	RW	RFIDPowerModel
RFID_ANT14_POWER	Antenna 14 Power	RW	RFIDPowerModel
RFID_ANT15_POWER	Antenna 15 Power	RW	RFIDPowerModel
RFID_ANT16_POWER	Antenna 16 Power	RW	RFIDPowerModel
KEY1	Pre set key 1	WO	array
KEY2	Pre set key 2	WO	array
KEY3	Pre set key 3	WO	array
KEY4	Pre set key 4	WO	array
KEY5	Pre set key 5	WO	array
KEY6	Pre set key 6	WO	array
KEY7	Pre set key 7	WO	array
KEY8	Pre set key 8	WO	array
KEY9	Pre set key 9	WO	array
KEY10	Pre set key 10	WO	array
KEY11	Pre set key 11	WO	array
KEY12	Pre set key 12	WO	array
KEY13	Pre set key 13	WO	array
KEY14	Pre set key 14	WO	array
KEY15	Pre set key 15	WO	array
KEY16	Pre set key 16	WO	array

8.1.3. UHF Class Parameter

Parameter Definition	Parameter Name	Read Write	ParameterType
UHF_PROPERTY	UHFAttribute	RO	UHFProperty
UHF_FREQUENCY	UHF Frequency Parameter	RW	UHFFrequencyModel
UHF_EPC_BASEBAND	UHF EPC Baseband Parameter	RW	UHFEPCBaseBandModel
UHF_EPC_ALGO	UHF EPC Algorithm Parameter	RW	UHFEPCAlgoModel
TAG_SELECT1	Tag Selection Rules1	RW	UHFSelectModel
TAG_SELECT2	Tag Selection Rules2	RW	UHFSelectModel
TAG_SELECT3	Tag Selection Rules3	RW	UHFSelectModel
TAG_SELECT4	Tag Selection Rules4	RW	UHFSelectModel
TAG_SELECT5	Tag Selection Rules5	RW	UHFSelectModel
TAG_SELECT6	Tag Selection Rules6	RW	UHFSelectModel
TAG_SELECT7	Tag Selection Rules7	RW	UHFSelectModel
TAG_SELECT8	Tag Selection Rules8	RW	UHFSelectModel

8.2. Parameter Definition

8.2.1. ArrayModel

Variable length array type Parameter

Attribute	Type	Description
Value	byte[]	Array content

8.2.2. BitmapModel

Bitmap type Parameter

Attribute	Type	Description
Value	bool[]	The content of the bitmap array can be set to its length through the constructor. (Used to set bitmap length before querying)

8.2.3. BoolModel

Bool type Parameter

Attribute	Type	Description
Value	bool	Parameter Value

8.2.4. ByteModel

Byte Type Parameter

Attribute	Type	Description
Value	byte	Parameter Value

8.2.5. BytesModel

Byte type fixed length array Parameter

Attribute	Type	Description
Value	byte[]	Parameter Value; The length of a fixed length array can be set through the constructor. (Fixed length array length needs to be set before querying)

8.2.6. DevicePwronModel

Device Power-on Parameter

Attribute	Type	Description
Uptime	int	Power-on time, unit: seconds
Reason	int	Reason for Power-on

8.2.7. FlashModel

Flash Parameter

Attribute	Type	Description
Enable	bool	Enable
WorkTime	short	Working duration, unit: ms
IdleTime	short	Idle duration, unit: ms

8.2.8. GPIOEventModel

GPIO Trigger Event Parameter

Attribute	Type	Description
RisingTrigger	bool	Enable rising edge triggering. Default value false
FallingTrigger	bool	Enable falling edge triggering. Default value false
Cmd	byte[]	Trigger instruction (complete instruction)

8.2.9. IntModel

Int Type Parameter

Attribute	Type	Description
Value	int	Parameter Value

8.2.10. MaskModel

Mask Type Parameter

Attribute	Type	Description

Attribute	Type	Description
Value	int	Parameter Value
Bit0	bool	Bit 0 Value
Bit1	bool	Bit 1Value
Bit2	bool	Bit 2Value
Bit3	bool	Bit 3Value
Bit4	bool	Bit 4Value
Bit5	bool	Bit 5Value
Bit6	bool	Bit 6Value
Bit7	bool	Bit 7Value
Bit8	bool	Bit 8Value
Bit9	bool	Bit 9Value
Bit10	bool	Bit 10Value
Bit11	bool	Bit 11Value
Bit12	bool	Bit 12Value
Bit13	bool	Bit 13Value
Bit14	bool	Bit 14Value
Bit15	bool	Bit 15Value
Bit16	bool	Bit 16Value
Bit17	bool	Bit 17Value
Bit18	bool	Bit 18Value
Bit19	bool	Bit 19Value
Bit20	bool	Bit 20Value
Bit21	bool	Bit 21Value
Bit22	bool	Bit 22Value
Bit23	bool	Bit 23Value
Bit24	bool	Bit 24Value
Bit25	bool	Bit 25Value
Bit26	bool	Bit 26Value
Bit27	bool	Bit 27Value
Bit28	bool	Bit 28Value
Bit29	bool	Bit 29Value
Bit30	bool	Bit 30Value
Bit31	bool	Bit 31Value

8.2.11. NetworkModel

Network Parameter

Attribute	Type	Description
DHCP	bool	Enable DHCP, default false
IP	IPAddress	IP Address, Default Value192.168.1.201
Mask	IPAddress	Subnet mask, Default Value255.255.255.0
Gateway	IPAddress	Gateway address, Default Value192.168.1.1
Mode	ENetworkMode	Network working mode
LocalPort	short	Local port, default 8090
RemotePort	short	Remote port number. Default Value8080
RemoteAddress	String	Remote IP address. Default Value192.168.1.99

8.2.12. PwrOnEventModel

Power-on trigger event Parameter

Attribute	Type	Description
Enable	bool	Enable startup trigger event. Default Valuefalse
Cmd	byte[]	Trigger instruction (complete instruction)

8.2.13. RFIDAntennaModel

天线配置 Parameter

Attribute	Type	Description
AntennaMask	uint	Work antenna mask，bit0-Ant1，bit1-Ant2...
SwitchTime	short	Antenna switching time (maximum dwell time)，unit: ms

8.2.14. RFIDEASEventModel

RFID EAS Event Parameter

Attribute	Type	Description
Enable	bool	启用 EAS 事件，默认 false
Source	short	标签数据源。可以为 EPC/UID、TID
StartBit	short	标签数据的起始位地址
Mask	byte[]	匹配掩码
Value	byte[]	匹配 Value
MatchedCmd	byte[]	匹配时触发的指令（完整指令）
MismatchedCmd	byte[]	不匹配时触发的指令（完整指令）

8.2.15. RFIDIdleModel

RFID Idle Parameter

Attribute	Type	Description
Enable	bool	Enable idle, default false
Period	short	Cycle, unit mA
IdlePercent	short	Idle percentage，0~100

8.2.16. RFIDKeyboardOutputModel

Simulated keyboard output configuration Parameter

Attribute	Type	Description
Enable	bool	Enable simulated keyboard output, default false
Source	short	Tag data source. Can be used for EPC/UID, TID
Format	byte	The keyboard output function converts the format of label data when sending it
EOF	byte[]	After the keyboard output function sends the label data, other data (in HEX format) such as 0A (enter), 09 (tab), and 20 (space) are sent. If it is empty, it

		means the None suffix.

8.2.17. RFIDPowerModel

RFID Antenna Power Parameter

Attribute	Type	Description
RdPower	short	Read power (inventory, label reading), unit: 0.01dBm
WrPowre	short	Write power (write label, lock label, destroy label), unit: 0.01dBm

8.2.18. RFIDPropertyModel

RFIDAttributeParameter

Attribute	Type	Description
SupportISO18K6B	bool	Supports or does not support IS18000-6B protocol
SupportISO18K6C	bool	Supports or does not support IS18000-6C protocol
SupportISO18K6D	bool	Supports or does not support IS18000-6D protocol
AntennaNumber	byte	Antenna Numbers
PowerMax	short	The maximum output power supported by the device, in units of 0.01dBm
PowerMin	short	The minimum output power supported by the device, in units of 0.01dBm
PowerStep	short	Device output power adjustment step value, unit: 0.01dBm
PowerDefault	short	Default output power of the device, unit: 0.01dBm

8.2.19. RFIDProtocolModel

RFID Protocal Parameter

Attribute	Type	Description
EnableISO18K6B	bool	Enable ISO18000-6B protocol
EnableISO18K6C	bool	Enable ISO18000-6C protocol
EnableISO18K6D	bool	Enable ISO18000-6D protocol

8.2.20. RFIDSpecialOutputModel

Special format output configuration Parameter

Attribute	Type	Description
Enable	bool	Enable special format output, default false
Source	short	Tag data source. Can be used for EPC/UID, TID
StartBit	short	Starting address of tag data
DataBits	short	Tag data bit length
Format	byte	When sending label data, perform format conversion on the data
CheckType	byte	Special format frame verification Type
AttachAntenna	bool	Attached antenna data
AttachRSSI	bool	Attached RSSI data
AttachFrequency	bool	Attached Frequency data
AttachPhase	bool	Attached phase data
AttachSN	bool	Attached SN
AttachAssertID	bool	Attached asset code
SOF	byte[]	Frame head
EOF	byte[]	Frame end
SEP	byte[]	Attached data delimiter

8.2.21. RFIDWiegandOuputModel

Wiegand output configuration Parameter

Attribute	Type	Description
Enable	bool	Enable Wigan output, default false
Source	short	Label data source. Can be used for

		EPC/UID, TID
StartBit	short	Starting address of tag data, -1 indicates the use of ending data
DataBits	short	Tag data bit length

8.2.22. SerialModel

Serial port configuration Parameter

Attribute	Type	Description
Bps	uint	Baud Rate，default 115200
DataBits	byte	Data bits, default 8
StopBits	byte	Stop position, default 1
Parity	ESerialParity	Check bit

8.2.23. ShortModel

Short Type Parameter

Attribute	Type	Description
Value	short	ParameterValue

8.2.24. StringModel

String Type Parameter

Attribute	Type	Description
Value	String	ParameterValue

8.2.25. TimestampModel

Time stamp Type Parameter

Attribute	Type	Description
Value	DateTime	ParameterValue

8.2.26. UHFEPCAlgoModel

UHF EPC Algorithm Parameter

Attribute	Type	Description
Type	EUHFAlgo	Algorithm Type
Param1	byte	Algorithm Parameter1
Param2	short	Algorithm Parameter2
Param3	int	Algorithm Parameter3

8.2.27. UHFEPCBaseBandModel

UHF EPC Baseband Parameter

Attribute	Type	Description
RFMode	byte	Work Mode
Session	byte	session
Target	byte	Target
QValueInit	byte	Initial QValue
QValueMax	byte	Max QValue
QValueMin	byte	Min QValue

8.2.28. UHFFrequencyModel

UHF Frequency Parameter

Attribute	Type	Description
Region	EUHFRegion	Work Area
HoppingMode	byte	Frequency hopping mode, 0 manual, 1 automatic
FrequencyChannelList	byte[]	channel list
FrequencyStartKHz	int	Starting frequency, unit: kHz
FrequencyStepKHz	short	Step frequency, unit: kHz
FrequencyChannelNumber	byte	Number of channels
FrequencyStart2KHz	int	Starting frequency 2, in kHz

FrequencyStep2KHz	short	Step frequency 2, unit kHz
FrequencyChannelNumber2	byte	Number of channels 2
HoppingTime	short	Frequency hopping time, unit mS, 0 is managed by the device itself

8.2.29. ParamModel

UHFAttributeParameter

Attribute	Type	Description
RFModeBitmap	bool[]	Work Mode Support Table
RegionBitmap	bool[]	Work Area Support Table

8.2.30. UHFSelectModel

UHF Tag Select Parameter

Attribute	Type	Description
IsOR	bool	Is it related to other Select or not. Default Valuefalse
Bank	EUHFBank	Select area, default ValueEPC
BitStart	short	Select the starting point of the matching bit
BitLen	short	Select matching bit length
MatchValue	byte[]	Select matching content

8.2.31. VersionInfoModel

Version Parameter

Attribute	Type	Description
Major	byte	Major version number
Branch	byte	Branch version number
Minor	byte	Sub version number
Patch	byte	Bug revision number
Descriptioin	String	Version description

8.2.32. WiFiAPModel

WiFi APParameter

Attribute	Type	Description
SSID	String	SSID for WiFi connection
Passwd	String	Password for WIFI connection

8.2.33. WiFiConnectModel

WiFi APParameter

Attribute	Type	Description
SSID	String	SSID for WiFi connection
Connected	bool	WIFI connection status

Appendix 1 UHF Work Area Table

The list of work areas supported by the device is determined by the factory of the device.

No.	Description	Frequency points
0	CMIIT 920~925MHz	16
1	FCC 902~928MHz	50
2	ETSI 866~867MHz	4
3	ETSI_HIGH 916~918MHz	3
4	ETSI_HYBRID 866~867MHz 916~918MHz	7
5	KOR 917~921MHz	6
6	JPN_NO_LBT 916~921MHz	4
7	JPN_LBT 916~921MHz	6
8	JPN_NO_LBT_19 916~924MHz	19
9	TWN 920~927MHz	14
10	MYS 919~923MHz	8
11	RUS_LOW 866~867MHz	5
12	RUS_HIGH 916~920MHz	4
13	AUS 920~926MHz	10
14	IND 865~867MHz	4
15	BRA 902~907.5MHz 915~928MHz	35
16	VIE 918~923MHz	6
17	ISR 915~917MHz	1
18	ZAF 916~920MHz	17
19	SGP 920~925MHz	10
20	THA 920~925MHz	10
240	TEST1 800~1000MHz	197
241	TEST2 840~960MHz	117
242	TEST3 860~960MHz	11
243	TEST4 860~930MHz	8